positive and real. So it would not make sense to identify $P(\lambda_i)$ with $\langle A|\lambda_i\rangle$. But the square of the magnitude, $\langle A|\lambda_i\rangle\langle\lambda_i|A\rangle$, is always positive and real and thus can be identified with the probability of a given outcome.

An important consequence of the principles is as follows:

The operators that represent observables are Hermitian.

The reason for this is twofold. First, since the result of an experiment must be a real number, the eigenvalues of an operator **L** must also be real. Secondly, the eigenvectors that represent unambiguously distinguishable results must have different eigenvalues, and must also be orthogonal. These conditions are sufficient to prove that **L** must be Hermitian.

3.3 AN EXAMPLE: SPIN OPERATORS

It may be hard to believe, but single spins—as simple as they are—still have a lot more to teach us about quantum mechanics, and we plan to milk them for all they're worth. Our goal in this section is to write down the spin operators in concrete form, as 2 × 2 matrices. Then, we'll get to see how they work in specific situations. We'll build up our spin operators and state-vectors shortly. But before we dive into the details, I'd like to say a little more about how operators are related to physical measurements. The relationship is a subtle one, and we'll say more about it as we go.

As you know, physicists recognize various types of physical quantities, such as scalars and vectors. It should come as no surprise, then, that an operator associated with the measurement of a vector (such as spin) has a vector character of its own.

In our travels so far, we have seen more than one kind of vector. The 3-vector is the most straightforward and serves as a prototype. It's a mathematical representation of an arrow in three-dimensional space, and is often represented by three real numbers, written out as a column matrix. Because their components are real-valued, 3-vectors are not quite rich